习题 4-3 解答

1.求下列不定积分.

(1) $\int x\sin 2x dx;$ (2) $\int \ln x dx;$

(3) $\int \sec^3 x dx;$ (4) $\int \sin(\ln x) dx;$

(5) $\int x^3 e^{-x^2} dx;$ (6) $\int e^{\sqrt{x}} dx;$

(7) $\int x\cos\frac{x}{2} dx;$ (8) $\int \frac{\ln\tan x}{\sin x\cos x} dx;$

(9) $\int x\tan^2 x dx;$ (10) $\int x^2 \arctan x dx;$

(11) $\int \frac{(\ln x)^3}{x^2} dx;$ (12) $\int e^{-2x}\sin\frac{x}{2} dx;$

(13) $\int x\sin x\cos x dx;$ (14) $\int x\cos^2 x dx;$

(15) $\int xe^{x^2}(1+x^2) dx;$ (16) $\int (arc\sin x)^2 dx;$

(17) $\int \frac{x\arctan x}{\sqrt{1+x^2}} dx;$ (18) $\int \frac{\arcsin\sqrt{x}}{\sqrt{1-x}} dx;$

(19) $\int \ln(1+x^2) dx;$ (20) $\int \frac{x}{\cos^2 x} dx;$

解：

$$(1)\int x\sin 2x\mathrm{d}x = \frac{1}{2}\int x\mathrm{d}(-\cos 2x) = -\frac{1}{2}x\cos 2x + \frac{1}{2}\int \cos 2x\mathrm{d}x$$

$$= -\frac{1}{2}x\cos 2x + \frac{1}{4}\sin 2x + C.$$

$$(2)\int \ln x\mathrm{d}x = x\ln x - \int x\mathrm{d}(\ln x) = x\ln x - \int x\frac{1}{x}\cdot \mathrm{d}x = x\ln|x| - x + C$$

(3) 因

$$\int \sec^3 x\mathrm{d}x = \int \sec x\cdot\sec^2 x\mathrm{d}x = \int \sec x\mathrm{d}(\tan x)$$

$$= \tan x\sec x - \int \tan^2 x\sec x\mathrm{d}x$$

$$= \tan x \sec x - \int \left(\sec^2 x - 1\right) \sec x \mathrm{d}x$$

$$= \tan x \sec x - \int \sec^3 x \mathrm{d}x + \int \sec x \mathrm{d}x$$

$$= \tan x \sec x + \ln\left|\sec x + \tan x\right| - \int \sec^3 x \mathrm{d}x.$$

故

$$\int \sec^3 x \mathrm{d}x = \frac{1}{2}\tan x \sec x + \frac{1}{2}\ln\left|\sec x + \tan x\right| + C.$$

(4) $\int \sin(\ln x)\mathrm{d}x = x\sin(\ln x) - \int x \mathrm{d}\left[\sin(\ln x)\right] = x\sin(\ln x) - \int \cos(\ln x)\mathrm{d}x$

$$= x\sin(\ln x) - x\cos(\ln x) + \int x \mathrm{d}\left[\cos(\ln x)\right]$$

$$= x\sin(\ln x) - x\cos(\ln x) - \int \sin(\ln x)\mathrm{d}x.$$

所以 $\int \sin(\ln x)\mathrm{d}x = \frac{x\sin(\ln x) - x\cos(\ln x)}{2} + C.$

(5) $\int x^3 \mathrm{e}^{-x^2}\mathrm{d}x = -\frac{1}{2}\int x^2 \mathrm{d}\mathrm{e}^{-x^2} = -\frac{1}{2}x^2\mathrm{e}^{-x^2} + \int x\mathrm{e}^{-x^2}\mathrm{d}x$，

$$= -\frac{1}{2}x^2\mathrm{e}^{-x^2} - \frac{1}{2}\int \mathrm{d}\mathrm{e}^{-x^2} = -\frac{1}{2}x^2\mathrm{e}^{-x^2} - \frac{1}{2}\mathrm{e}^{-x^2} + C.$$

(6) 令 $\sqrt{x} = u$，则 $x = u^2$，$\mathrm{d}x = 2u\mathrm{d}u$，于是

$$\int \mathrm{e}^{\sqrt{x}}\mathrm{d}x = \int \mathrm{e}^u 2u\mathrm{d}u = \int 2u\mathrm{d}(\mathrm{e}^u) = 2u\mathrm{e}^u - 2\int \mathrm{e}^u \mathrm{d}u = 2u\mathrm{e}^u - 2\mathrm{e}^u + C$$

$$= 2\sqrt{x}\mathrm{e}^{\sqrt{x}} - 2\mathrm{e}^{\sqrt{x}} + C.$$

(7) $\int x\cos\frac{x}{2}dx = 2\int x d\sin\frac{x}{2} = 2x\sin\frac{x}{2} - 4\int \sin\frac{x}{2}d\frac{x}{2} = 2x\sin\frac{x}{2} + 4\cos\frac{x}{2} + C;$

$(9)\int x\tan^2 x dx = \int x\left(\sec^2 x - 1\right)dx = \int x\sec^2 x dx - \int x dx$

$$= \int x d\tan x - \int x dx = x\tan x - \int \tan x dx - \int x dx$$

$$= x\tan x + \ln\left|\cos x\right| - \frac{1}{2}x^2 + C;$$

(10) $\int x^2 \arctan x\mathrm{d}x = \int \arctan x \mathrm{d}\left(\frac{1}{3}x^3\right) = \frac{1}{3}x^3\arctan x - \frac{1}{3}\int \frac{x^3}{1+x^2}\mathrm{d}x$

$$= \frac{1}{3}x^3\arctan x - \frac{1}{3}\int\left(x - \frac{x}{1+x^2}\right)\mathrm{d}x$$

$$= \frac{1}{3}x^3\arctan x - \frac{1}{3}\int x\mathrm{d}x + \frac{1}{3}\int \frac{x}{1+x^2}\mathrm{d}x$$

$$= \frac{1}{3}x^3\arctan x - \frac{1}{6}x^2 + \frac{1}{6}\ln\left(1+x^2\right) + C.$$

$$(11)\int\frac{(\ln x)^3}{x^2}dx=-\int(\ln x)^3\,d\frac{1}{x}=-\frac{(\ln x)^3}{x}+\int\frac{1}{x}3(\ln x)^2\frac{1}{x}dx$$

$$=-\frac{(\ln x)^3}{x}+\int\frac{1}{x^2}3(\ln x)^2\,dx=-\frac{(\ln x)^3}{x}-3\int(\ln x)^2\,d\frac{1}{x}$$

$$=-\frac{(\ln x)^3}{x}-\frac{3(\ln x)^2}{x}+3\int\frac{1}{x}d(\ln x)^2=-\frac{(\ln x)^3}{x}-\frac{3(\ln x)^2}{x}+3\int\frac{2\ln x}{x^2}dx$$

$$=-\frac{(\ln x)^3}{x}-\frac{3(\ln x)^2}{x}-6\int\ln x d\frac{1}{x}$$

$$=-\frac{(\ln x)^3}{x}-\frac{3(\ln x)^2}{x}-\frac{6\ln x}{x}+6\int\frac{1}{x^2}dx$$

$$=-\frac{(\ln x)^3}{x}-\frac{3(\ln x)^2}{x}-\frac{6\ln x}{x}-\frac{6}{x}+C.$$

(12) 因

$$\int e^{-2x}\sin\frac{x}{2}dx=-\frac{1}{2}\int\sin\frac{x}{2}d\left(e^{-2x}\right)=-\frac{1}{2}e^{-2x}\sin\frac{x}{2}+\frac{1}{2}\int e^{-2x}\cdot\frac{1}{2}\cos\frac{x}{2}dx$$

$$=-\frac{1}{2}e^{-2x}\sin\frac{x}{2}-\frac{1}{8}\int\cos\frac{x}{2}d\left(e^{-2x}\right)$$

$$=-\frac{1}{2}e^{-2x}\sin\frac{x}{2}-\frac{1}{8}e^{-2x}\cos\frac{x}{2}+\frac{1}{8}\int e^{-2x}\cdot\left(-\frac{1}{2}\right)\sin\frac{x}{2}dx$$

$$=-\frac{1}{8}e^{-2x}\left(4\sin\frac{x}{2}+\cos\frac{x}{2}\right)-\frac{1}{16}\int e^{-2x}\sin\frac{x}{2}dx.$$

故

$$\int e^{-2x}\sin\frac{x}{2}dx=-\frac{2}{17}e^{-2x}\left(4\sin\frac{x}{2}+\cos\frac{x}{2}\right)+C.$$

（13） $\int x\sin x\cos x dx=\frac{1}{2}\int x\sin 2x dx=\int x d\left(-\frac{1}{4}\cos 2x\right)$

$$=-\frac{1}{4}x\cos 2x+\frac{1}{4}\int\cos 2x dx=-\frac{1}{4}x\cos 2x+\frac{1}{8}\sin 2x+C.$$

$$(14)\int x\cos^2 xdx=\int x\frac{1+\cos 2x}{2}dx=\frac{1}{2}\int x(1+\cos 2x)dx=\frac{1}{2}\int xdx+\frac{1}{2}\int x\cos 2xdx$$

$$=\frac{1}{2}\int xdx+\frac{1}{4}\int xd\sin 2x=\frac{1}{2}\int xdx+\frac{1}{4}x\sin 2x-\frac{1}{4}\int\sin 2xdx$$

$$=\frac{1}{4}x^2+\frac{1}{4}x\sin 2x+\frac{1}{8}\cos 2x+C.$$

(15) $\int xe^{x^2}(1+x^2)dx=\int e^{x^2}(1+x^2)\cdot xdx=\frac{1}{2}\int e^{x^2}(1+x^2)d\left(x^2\right)$. 令 $x^2=t$，则

$$\int x\mathrm{e}^{x^2}(1+x^2)\mathrm{d}x=\frac{1}{2}\int \mathrm{e}^{x^2}(1+x^2)\,\mathrm{d}\left(x^2\right)=\frac{1}{2}\int \mathrm{e}^{t}(1+t)\mathrm{d}t=\frac{1}{2}\mathrm{e}^{t}+\frac{1}{2}\int t\mathrm{e}^{t}\mathrm{d}t$$
$$=\frac{1}{2}\mathrm{e}^{t}+\frac{1}{2}\int t\mathrm{d}\mathrm{e}^{t}=\frac{1}{2}\mathrm{e}^{t}+\frac{1}{2}t\mathrm{e}^{t}-\frac{1}{2}\mathrm{e}^{t}+C$$
$$=\frac{1}{2}x^2\mathrm{e}^{x^2}+C.$$

(16) $$\int(\arcsin x)^2\mathrm{d}x=x(\arcsin x)^2-\int x\mathrm{d}(\arcsin x)^2$$
$$=x(\arcsin x)^2-\int 2\arcsin x\cdot\frac{x}{\sqrt{1-x^2}}\mathrm{d}x$$
$$=x(\arcsin x)^2+2\int \arcsin x\mathrm{d}\left(\sqrt{1-x^2}\right)$$
$$=x(\arcsin x)^2+2\sqrt{1-x^2}\arcsin x-2\int\frac{\sqrt{1-x^2}}{\sqrt{1-x^2}}\mathrm{d}x$$
$$=x(\arcsin x)^2+2\sqrt{1-x^2}\arcsin x-2x+C.$$

(17) $$\int\frac{x\arctan x}{\sqrt{1+x^2}}\mathrm{d}x=\int\arctan x\cdot\frac{x}{\sqrt{1+x^2}}\mathrm{d}x=\int\arctan x\mathrm{d}\sqrt{1+x^2}$$
$$=\sqrt{1+x^2}\arctan x-\int\frac{1}{\sqrt{1+x^2}}\mathrm{d}x$$
$$=\sqrt{1+x^2}\arctan x-\ln\left(x+\sqrt{1+x^2}\right)+C.$$

$$(18)\int\frac{\arcsin\sqrt{x}}{\sqrt{1-x}}dx=\int\frac{\arcsin\sqrt{x}}{\sqrt{1-\left(\sqrt{x}\right)^2}}dx=2\int\sqrt{x}\arcsin\sqrt{x}d\arcsin\sqrt{x}$$
$$=\int\sqrt{x}d\left(\arcsin\sqrt{x}\right)^2=\sqrt{x}\left(\arcsin\sqrt{x}\right)^2-\int\left(\arcsin\sqrt{x}\right)^2d\sqrt{x}$$
$$=\sqrt{x}\left(\arcsin\sqrt{x}\right)^2-\sqrt{x}\left(\arcsin\sqrt{x}\right)^2-2\sqrt{1-x}\arcsin\sqrt{x}+2\sqrt{x}+C$$
$$=-2\sqrt{1-x}\arcsin\sqrt{x}+2\sqrt{x}+C.$$

(19) $$\int\ln(1+x^2)\mathrm{d}x=x\ln\left(1+x^2\right)-2\int\frac{x^2}{1+x^2}\mathrm{d}x=x\ln\left(1+x^2\right)-2\int\frac{\left(1+x^2\right)-1}{1+x^2}\mathrm{d}x$$
$$=x\ln\left(1+x^2\right)-2x+2\arctan x+C.$$

$$(20)\int\frac{x}{\cos^2 x}dx=\int x\sec^2 x\mathrm{d}x=\int x\mathrm{d}\left(\tan x\right)=x\tan t x-\int\tan x\mathrm{d}x$$
$$=x\tan x+\ln\left|\cos x\right|+C.$$

2.证明下列递推公式.

(1) $I_n=\int(\ln x)^n\,dx=x(\ln x)^n-nI_{n-1}$ (n 为正整数);

(2) $I_n=\int\cos^n x dx=\dfrac{\cos^{n-1}x\sin x}{n}+\dfrac{n-1}{n}I_{n-2}$ (n 为正整数);

证 (1)

$$I_n=\int(\ln x)^n\,dx=x(\ln x)^n-\int x d(\ln x)^n=x(\ln x)^n-n\int(\ln x)^{n-1}dx=x(\ln x)^n-nI_{n-1}.$$

（2）由于

$$\begin{aligned}I_n&=\int\cos^n x\mathrm{d}x=\int\cos^{n-1}x\cdot\cos x\mathrm{d}x=\int\cos^{n-1}x\mathrm{d}(\sin x)\\&=\cos^{n-1}x\cdot\sin x-\int\sin x\mathrm{d}(\cos^{n-1}x)\\&=\cos^{n-1}x\cdot\sin x+(n-1)\int\sin^2x\cos^{n-2}x\mathrm{d}x\\&=\cos^{n-1}x\cdot\sin x+(n-1)\int(1-\cos^2x)\cos^{n-2}x\mathrm{d}x\\&=\cos^{n-1}x\cdot\sin x+(n-1)\int\cos^{n-2}x\mathrm{d}x-(n-1)\int\cos^n x\mathrm{d}x\\&=\cos^{n-1}x\cdot\sin x+(n-1)I_{n-2}-(n-1)I_n\end{aligned}$$

所以 $I_n=\dfrac{\cos^{n-1}x\cdot\sin x}{n}+\dfrac{(n-1)}{n}I_{n-2}$.

3.已知 $f(x)$ 的一个原函数为 e^{-x^2}, 求 $\int xf'(x)dx$.

解 依题意 $\int f(x)\mathrm{d}x=\mathrm{e}^{-x^2}+C$, 且 $f(x)=\left(\mathrm{e}^{-x^2}\right)'=-2x\mathrm{e}^{-x^2}$, 故

$$\int xf'(x)dx=\int xdf(x)=xf(x)-\int f(x)dx=-2x^2\mathrm{e}^{-x^2}-\mathrm{e}^{-x^2}+C.$$